

分为实部和虚部 —— 提取半角 $\rightarrow$ 对称比较巧

核心思想: 把一个“不对称”的表达式强行变成一个“对称”的表达式.

$$\begin{cases} e^{i\theta} + e^{-i\theta} = 2\cos\theta & (\text{Re}) \\ e^{i\theta} - e^{-i\theta} = 2i\sin\theta & (\text{Im}) \end{cases}$$

① $e^{i\theta} + 1$

$$\Rightarrow e^{i\frac{\theta}{2}} (e^{i\frac{\theta}{2}} + e^{-i\frac{\theta}{2}})$$

$$= e^{i\frac{\theta}{2}} \cdot 2\cos\frac{\theta}{2}$$

$$= (\underbrace{\cos\frac{\theta}{2}}_{\text{Re}} + i \underbrace{\sin\frac{\theta}{2}}_{\text{Im}}) \cdot (2\cos\frac{\theta}{2})$$

② $e^{i\theta} - 1$

$$\Rightarrow e^{i\frac{\theta}{2}} (e^{i\frac{\theta}{2}} - e^{-i\frac{\theta}{2}})$$

$$= e^{i\frac{\theta}{2}} \cdot 2i\sin\frac{\theta}{2}$$

$$= (\underbrace{\cos\frac{\theta}{2}}_{\text{Re}} + i \underbrace{\sin\frac{\theta}{2}}_{\text{Im}}) \cdot (2i\sin\frac{\theta}{2})$$

③ $e^{iA} + e^{iB}$

$$\Rightarrow e^{i(\frac{A+B}{2})} (e^{i(\frac{A-B}{2})} + e^{i(\frac{B-A}{2})})$$

$$= e^{i\frac{A+B}{2}} (2\cos\frac{A-B}{2})$$

④ $e^{iA} - e^{iB}$

$$\Rightarrow e^{i(\frac{A+B}{2})} (e^{i(\frac{A-B}{2})} - e^{i(\frac{B-A}{2})})$$

$$= e^{i(\frac{A+B}{2})} (2i\sin\frac{A-B}{2})$$

Pf: Prove $1 + \cos\theta + \cos 2\theta + \cos 3\theta + \dots + \cos n\theta = \frac{1}{2} + \frac{\sin[(n+\frac{1}{2})\theta]}{2\sin(\frac{\theta}{2})}$

$$1 + \cos\theta + \cos 2\theta + \dots + \cos n\theta = \text{Re} (e^{i0} + e^{i\theta} + e^{i2\theta} + \dots + e^{in\theta})$$

$$= \text{Re} \left(\frac{e^{i(n+1)\theta} - 1}{e^{i\theta} - 1} \right)$$

$$e^{i(n+1)\theta} - 1 = e^{i(\frac{n+1}{2})\theta} (e^{i(\frac{n+1}{2})\theta} - e^{-i(\frac{n+1}{2})\theta})$$

$$= e^{i(\frac{n+1}{2})\theta} \cdot 2i\sin(\frac{n+1}{2})\theta$$

$$e^{i\theta} - 1 = e^{i\frac{\theta}{2}} (e^{i\frac{\theta}{2}} - e^{-i\frac{\theta}{2}})$$

$$= e^{i\frac{\theta}{2}} \cdot 2i \sin \frac{\theta}{2}$$

$$\frac{e^{i(n+1)\theta} - 1}{e^{i\theta} - 1} = \frac{e^{i(\frac{n+1}{2})\theta} \sin(\frac{n+1}{2})\theta}{e^{i\frac{\theta}{2}} \sin \frac{\theta}{2}}$$

$$= e^{i\frac{n}{2}\theta} \frac{\sin(\frac{n+1}{2})\theta}{\sin \frac{\theta}{2}}$$

$$= \left(\cos \frac{n\theta}{2} + i \sin \frac{n\theta}{2} \right) \frac{\sin(\frac{n+1}{2})\theta}{\sin \frac{\theta}{2}}$$

$$\operatorname{Re} \left(\frac{e^{i(n+1)\theta} - 1}{e^{i\theta} - 1} \right) = \frac{\sin(\frac{n+1}{2})\theta \cos \frac{n\theta}{2}}{\sin \frac{\theta}{2}} = \frac{\frac{1}{2} (\sin[(n+\frac{1}{2})\theta] + \sin \frac{\theta}{2})}{\sin \frac{\theta}{2}}$$

$$= \frac{1}{2} + \frac{\sin((n+\frac{1}{2})\theta)}{2 \sin \frac{\theta}{2}}$$